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Applied Artificial Intelligence
An International Journal

ISSN: 0883-9514 (Print) 1087-6545 (Online) Journal homepage: www.tandfonline.com/journals/uaai20

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To cite this article: Sergio E. Calizaya-Milla, Jair Santos & Fredy Abel Huanca Torres (2026) Fine-Tuning Monolingual Pre-Trained BERT Models for Sentiment Analysis in Peruvian Slang Contexts, Applied Artificial Intelligence, 40:1, 2641381, DOI: 10.1080/08839514.2026.2641381

To link to this article: https://doi.org/10.1080/08839514.2026.2641381

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Published online: 14 Mar 2026.

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APPLIED ARTIFICIAL INTELLIGENCE
2026, VOL. 40, NO. 1, e2641381 (15 pages)
https://doi.org/10.1080/08839514.2026.2641381

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Fine-Tuning Monolingual Pre-Trained BERT Models for Sentiment Analysis in Peruvian Slang Contexts

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ABSTRACT

The continuous advancements in natural language processing (NLP) have led to the development of highly effective models such as BERT, RoBERTa, GPT-4, Llama 3, and Gemini. However, adapting these models to specific dialects, especially in languages other than English, remains underexplored, particularly in slang or informal language. Our research evaluates monolingual Spanish models that best fit Peruvian colloquial expressions in response to this need. Our approach involved constructing a specialized dataset of 11,276 manually annotated social media comments, preprocessed to retain the unique features of Peruvian slang. We also expanded the models' vocabulary using a dictionary of Peruvian slang, ensuring better recognition of local expressions. The dataset was used to fine-tune the models, with RoBERTuito demonstrating superior performance, achieving an F1-score of 0.750, significantly outperforming BETO (0.661), BERTuit (0.700), and RoBERTa-BNE (0.696). This research offers a robust solution for sentiment analysis in Peruvian Spanish. It sets a benchmark for adapting monolingual models to linguistic contexts, with applications extending to other dialects and informal language variants.

ARTICLE HISTORY
Received 7 July 2024
Revised 20 March 2025
Accepted 23 March 2025

Introduction

The field of natural language processing (NLP) has advanced significantly in recent years, allowing for the creation of models based on Transformers such as BERT, RoBERTa, GPT-4o, Llama 3, and Gemini, among others, trained with hundreds of millions of parameters. These models have demonstrated a strong ability to process natural language and provide information effectively. Large language models (LLMs) are mostly pre-trained on large datasets in English and other languages, aiming for the model to learn generic or high-level functions that can be transferred and fine-tuned for specific tasks. LLMs can perform various NLP tasks, including translating text from one language to another, generating, summarizing, classifying text, answering questions, and more. In the case of the Spanish language, multilingual models from major private companies have shown better results in specific tasks than monolingual models (Agerri and Agirre 2023). As a result, it initially seemed unnecessary to dedicate resources to building models specifically for Spanish. However, there is no guarantee that major companies will keep their models updated, which may cause them to become obsolete very quickly. Furthermore, case studies have shown that models like multilingual BERT prefer sentences similar to English (Papadimitriou, Lopez, and Jurafsky 2022). Multilingual BERT has been outperformed in other languages by monolingual models, such as FinBERT (Virtanen et al. 2019), a model based on BERT trained with three different Finnish language text corpora.

It has been demonstrated that superior classification accuracies can be achieved when BERT is complemented with an effective preprocessing method. An example of this is observed in the Italian language, where sentiment analysis of tweets in this language has shown positive results (Pota et al. 2021). This preprocessing strategy could be applicable and further explored in other languages using pre-trained models.

In this context, the impact of corpus size, quality, and pre-training techniques must be examined to ensure that monolingual Spanish models significantly outperform multilingual models (Agerri and Agirre 2023). As a result of this challenge, monolingual models in Spanish have emerged, such as BETO, a BERT-based model

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for Spanish (Cañete et al. 2023), TwilBERT, a BERT adaptation trained with Spanish tweets (González, Hurtado, and Pla 2021), MarIA (Gutiérrez-Fandiño et al. 2022), a RoBERTa-based language model developed with a massive corpus of 570 GB of text from the National Library of Spain, RigoBERTa (Serrano et al. 2022), RoBERTuito (Pérez et al. 2022), based on the RoBERTa architecture and also trained with tweets, ALBETO and DistilBETO (Cañete et al. 2022), based on the ALBERT and DistilBERT architectures for the Spanish language, which has achieved better results compared to base multilingual models.

Most sentiment analysis work in Latin America is based on lexicons with polarity. Applications such as SAET (Utitiaj, Morillo, and Vallejo 2020), in an Ecuadorian context, demonstrate similar performance compared to commercial sentiment analysis tools. However, as lexicon-based models, they do not achieve high accuracy because they fail to account for additional parameters, such as the grammatical structures of the Spanish language. In Peru, as of this study, no comparable studies have been identified that apply NLP using monolingual Transformer architecture models to analyze colloquial language, including local slang, within this same context or approach.

Sentiment analysis is useful for understanding opinions on various topics such as politics, entertainment, sports, and local products and services. However, its accuracy in Peruvian Spanish is limited by the language's unique features, cultural nuances, and the challenges of working with slang, which is often perceived as noise and may be excluded from the corpus (Bhattacharyya, Dhuliawala, and Kanojia 2016). Furthermore, ambiguity caused by abbreviations and the lack of context complicates interpretation (Teodorescu and Saharia 2015). Slang, on the other hand, is not usually part of a language's standard vocabulary (Gómez-Adorno et al. 2016), making its recognition difficult. Additionally, the meaning of slang can vary between positive and negative depending on the context in which it is used (Mao, Liu, and Zhang 2024). Therefore, text processing models require specific dictionaries for each language studied (Gómez-Adorno et al. 2016), and adapting a Transformer model to Peruvian Spanish could enhance both understanding and accuracy in sentiment analysis. This research addresses how to adapt monolingual Transformer models to optimize their effectiveness in the Peruvian context, evaluating their performance in natural language processing.

The sections of this study are organized as follows: in Related Work, we review the state of the art and relevant models, contextualizing the need to adapt a language model to Peruvian slang. Next, we provide a detailed description of the methodological process, including dataset construction, preprocessing techniques, data annotation, and model fine-tuning. In Results, we present the quantitative evaluation of the model's performance and the obtained results. In the Discussion, we examine the applicability of the findings and their impact on sentiment analysis in specific linguistic contexts. Finally, in Conclusion, we summarize the study's main contributions and outline future research directions.

Related works

The task of text classification within sentiment analysis has become an active research field in natural language processing (Pota et al. 2021). It has applications in various areas such as marketing, politics, customer service, and tourism. Similarly, it is relevant for professionals and researchers who seek to study human-computer interaction (Hutto and Gilbert 2014).

In natural language processing, different techniques are applied for text polarity classification. Among the traditional approaches is the lexicon-based approach, an advantageous technique for sentiment analysis (Hutto and Gilbert 2014). It uses a lexicon dictionary that assigns a polarity (positive, negative, or neutral) to each term or expression. We also find deep learning algorithms commonly used in this field, such as Long Short-Term Memory (LSTM), Recurrent Neural Networks (RNN), and Convolutional Neural Networks (CNN) (Wu et al. 2023).

Transformer architecture, particularly BERT-based models, currently represents one of the most advanced techniques in natural language processing. These models are trained on large amounts of textual data and then fine-tuned for specific tasks, such as text polarity classification (Karfi and Fkihi 2022). Transformer-based models have achieved state-of-the-art results in many natural language processing tasks. On the other hand, Pota et al. (2021) affirm that bidirectional encoding Transformer models achieve outstanding results in text recognition and classification.

The TwilBERT model (González, Hurtado, and Pla 2021), derived from BERT and specialized in the Spanish language within the Twitter domain, tackles 14 text classification tasks using a response order predictor to learn coherence in each tweet. In this way, it outperforms the multilingual version of BERT by up to +11.07 in the F1-Score metric. This research provides a framework using the Keras library to train and fine-tune the TwilBERT model.

On the other hand, RoBERTuito (Pérez et al. 2022), a Transformer model that uses the RoBERTa and BERTweet architectures, is trained on Spanish tweets and is considered by the authors to be competitive in multilingual tasks. Five hundred million Spanish tweets and specific datasets, such as regional or thematic variants, were used for the training. The model's weights are published on the HuggingFace platform for free use. The average F1-Score results, compared to other models like RoBERTa and BETO, surpass them in sentiment analysis and hate speech detection tasks.

Training a model from scratch, such as BERT and its Spanish version BETO, requires significant computational resources and time. To avoid this, lightweight Transformer models with reduced parameters, such as ALBERT (Lan et al. 2020) and DistilBERT (Jiao et al. 2020), have been created, though they are trained in English. Cañete et al. (2022) propose ALBETO and DistilBETO, models pre-trained exclusively on Spanish corpora. The results of these models are on par with the BETO model. The authors have made the models freely available for future research.

Various studies have created pre-trained Transformer models derived from BERT to train them in a specific language. CamemBERT (Martin et al. 2020) and FlauBERT (Le et al. 2020) for the French language, RoBERT (Delobelle, Winters, and Berendt 2020) for Dutch and FinBERT (Virtanen et al. 2019), for Finnish. Additionally, the IndoBERT model in the Indonesian language is fine-tuned for the recognition of slang in that language (Fernandez et al. 2022). All these pre-trained models have proven to be better than multilingual models.

Despite the numerous previous studies that have applied Transformer-based models to classify text polarity, most of these studies have focused on English texts, and very few address Spanish (Utitiaj, Morillo, and Vallejo 2020). This has revealed a notable lack of models dedicated to polarity classification in Spanish texts, especially in the Peruvian colloquial language.

Methodological process

This explanatory research focuses on a less-explored aspect of sentiment analysis: adapting monolingual language models to a dataset containing Peruvian Spanish slang. Although sentiment analysis has been widely studied in other contexts and languages, no specific studies have been found to address the particularities of Peruvian Spanish and its colloquial expressions. Therefore, our work seeks to fill this gap for dialectal variants, focusing on Peruvian slang.

The present study began with the extraction of texts from comments on the Facebook platform, using Meta's Graph API through the open-source software Facepager.[1] This initial stage allowed the formation of a corpus in Spanish. Subsequently, the corpus underwent a rigorous cleaning and filtering process, where a specialized dictionary of Peruvian slang was applied to identify and exclude comments that contained slang and colloquial expressions. Once the corpus was refined, we built the dataset through a manual annotation process. This dataset was then used to perform fine-tuning (full fine-tuning) of four pre-trained models in Spanish: RoBERTuito, BETO, BERTuit, and RoBERTa-BNE. Finally, the fine-tuned models were tested on the text classification task for sentiment analysis, focusing on determining the polarity of comments as positive, negative, or neutral. Standard classification metrics, precision, recall, accuracy, and F1-score, were used to evaluate the models' performance. These steps are detailed in Figure 1.

Dataset construction

The study population consisted of comments posted by Facebook users in Peru. Data collection was carried out using Meta's Graph API and Facepager. We began collecting comments based on the methodology and evaluation of a study on Peruvian influencers conducted by Perú (2023), which considered the number of followers, interaction, impact, and influence on social networks. These comments cover entertainment, technology, gaming, news, sports, fashion, gastronomy, lifestyle,

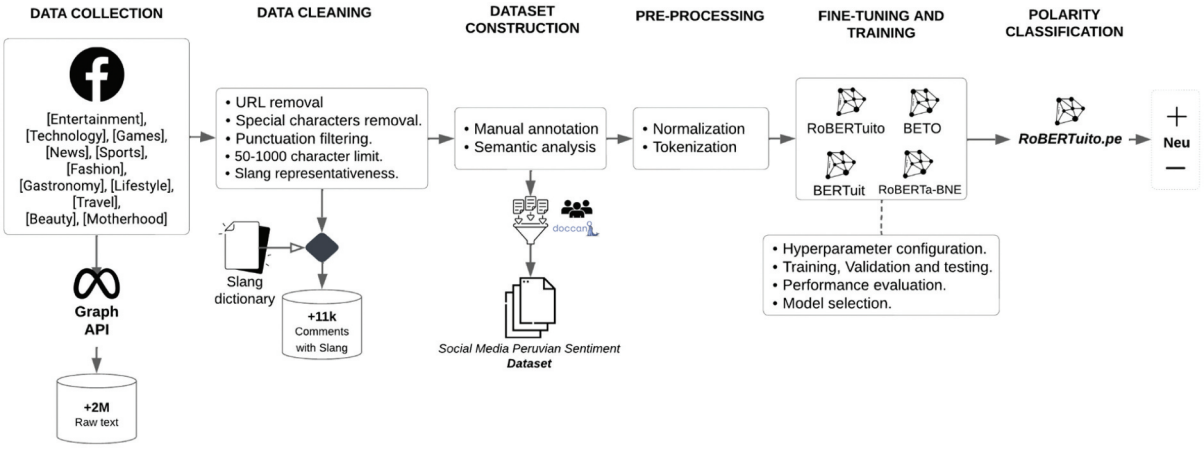


Figure 1. Diagram of the proposed fine-tuning method for BERT models.

Table 1. Distribution of thematic domains in the dataset.

Thematic domains	%
Sport	13%
Entertainment	12%
Gamers	8%
Gastronomy/Foodies	13%
Maternity	13%
Fashion/Beauty/Lifestyle	13%
News	6%
Popular/Showbiz	12%
Technology	3%
Trips	7%
Total	**100%**

celebrity news, travel, beauty, and motherhood. The data collection spanned five years (2019–2023), resulting in a total of 2.2 million comments gathered. The distribution of these thematic domains is summarized in Table 1.

To construct the final dataset, a cleaning and filtering process was conducted on the initial 2.2 million comments, and following the study by (Cuzcano, Cuzcano, and Ayma 2020), we considered including comments with emojis to enhance the emotional content of the messages. This procedure resulted in a dataset with over 90,000 comments, each characterized by including slang identified through our slang dictionary. To obtain a preliminary insight, we decided to select a representative sample of these comments. For this, we applied a sampling technique based on the determination of sample size in finite populations (Montesinos et al. 2010).

$$n = \frac{NZ^2p(1-p)}{(N-1)E^2 + Z^2p(1-p)} \tag{1}$$

Where:

- N: 90,000, the population size
- Z: 2.58, the 99% confidence level
- p: 0.5, estimated proportion of the characteristic of interest
- E: 0.01, margin of error

As a result, we obtained around 14,044 comments. After cleaning and refining, the final sample was reduced to 11,276 valid comments, which formed our final dataset, SocialMediaPeruvianSentiment (SMPS). This procedure allowed us to obtain a diverse yet manageable dataset for further analysis.

For comment filtering, we built a dictionary of slang gathered from forums, social media, chats, websites, and the dictionary of Americanisms from the Association of Academies of the Spanish Language ASALE.[2] This Peruvian slang dictionary[3] is available for free download.

Preprocessing

The text cleaning and normalization process included the removal of noise, such as special characters and typographical errors. Techniques like stemming or lemmatization were avoided to preserve the semantic structure of the slang and ensure its correct interpretation. To guarantee the quality and consistency of the dataset, the following procedures were implemented:

- Standardization of uppercase and lowercase usage: All text was converted to lowercase to ensure uniformity and minimize discrepancies, reducing corpus variability. This standardization simplifies the vocabulary and decreases the number of unique words, thus improving the performance of classification models.
- Removal of accents: Accents and non-ASCII characters, except for emojis, were removed due to their inconsistent use of informal language. Some classification models experience slight performance improvements when the text is processed without accents.
- Reduction of repetitive characters: To mitigate excessive repetition in words like "Paltaaaaaaaaaaaa," which was normalized to "Paltaaa," and "Causaaaaaaa," which was shortened to "Causaaa," a limit of three consecutive characters was set.
- Standardization of laughter expressions: Laughter expressions like "jajajajajajja" or "hahaha" were converted to "jajaja," standardizing their format to maintain consistency in the dataset.
- Emoji standardization: For the RoBERTuito model, emojis were converted to textual representations, for example, "🤮" to "vomiting face emoji" or "😊" to "smiling face emoji," since the base model is trained this way. For this, we used the emoji[4] library. For the other models, emojis were kept in hexadecimal format. Removing emojis for all models was considered, but it did not significantly improve performance.
- Removal of mentions, hashtags, and links: User mentions were replaced with "@username," hashtags were normalized by keeping only the text content (for example, "#BreakingNews" became "breaking news"), and URLs were removed entirely. This process contributed to the dataset's consistency, preserved user anonymity, and eliminated irrelevant data contained in the URLs. However, in some cases, mentions on Facebook included partial or complete names, allowing those comments to be identified as invalid during the annotation process. For example, "Juan Andres es tu jato jajaja." Due to informal language, these mentions were not recognized as specific tags or links. Instead, the relevance of the comment was evaluated based on its content and length, which allowed for filtering out comments that did not provide significant value to the analysis.

A specialized Peruvian slang dictionary was essential to identify cases where words contained slang embedded in their lexical structure. For example, in the comment "Yo mirando todo el vídeo y diciendo ahorita le da una patada a la pared" ("I was watching the whole video and saying he is about to kick the wall"), the word "patada" (kick) encapsulates the slang term "pata," which in Peruvian slang is interpreted as "friend." Similarly, polysemous terms were detected in numerous comments. For example, in "mi perro también se lastimó la pata . . ." ("my dog also hurt its paw . . ."), the term "pata" is understood in its literal sense. However, in a different context, such as "Yo concuerdo en muchas cosas con el pata . . ." ("I agree with many things the guy says . . ."), the same term "pata" takes on its colloquial meaning, referring to a friend. This observation highlights the complexity and semantic richness of colloquial language and slang, which can refer to more than one meaning at once (Teodorescu and Saharia 2015).

To address this difficulty, we implemented an additional validation based on cosine similarity (Kwartler 2017); the general formula (2) results in the cosine of the angle θ between two vectors, which ranges from 0 to 1. A cosine similarity close to 1 indicates that the word and the slang are highly aligned semantically, while a value close to 0 indicates little or no similarity.

$$Cosine\ Similarity = \frac{\vec{A} \cdot \vec{B}}{\| \vec{A} \| \cdot \| \vec{B} \|} = \frac{\sum_{i=1}^{n} A_i \cdot B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \cdot \sqrt{\sum_{i=1}^{n} B_i^2}} \quad (2)$$

During this validation, we observed that 91% of the comments in the dataset have a cosine similarity greater than 0.7, indicating high accuracy in identifying slang within the comments. Table 2 shows some examples of the calculation for each identified slang.

In Figure 2, an N-Gram highlights the predominant terms in the analyzed dataset. This visual representation displays the frequency with which certain slang and colloquial expressions appear in the comments.

Annotation process

For annotating the comments, we used Doccano,[5] an open-source tool with collaborative text labeling functions. The annotation process involved a review by eight native Peruvians, aged between 18 and 40, who were given access to the Doccano tool to annotate the comments randomly, where 1 = positive, 2 = neutral, and 3 = negative (see Table 3). Once the annotation of the comments was completed, we obtained the polarity distribution (see Table 4). Similarly, Figure 4 shows the most frequent slang terms in each positive, negative, and neutral class. The SMPS dataset has been publicly shared on Hugging Face.[6]

Each annotator reviewed the comments randomly, and these were labeled into three categories: positive (1), neutral (2), and negative (3). Below, we clearly define each of these polarities:

Table 2. Examples of cosine similarity between identified slang and recognized words.

Text	Slang	Word	Cosine Similarity
Joel Apolaya broder 😅 😅	broder	Joel	0.292
Literal fría yara 🤣 🤣 🤣	yara	fría	0.325
Cholo. En Premios la Zona estabas Bien Charlie … 😎 😎 😂	charli	Charlie …	0.444
Te quedo de lujo! Me has dado varias ideas de como darle mas sabor a mi chaufita, Gracias! 😄 👌	chau	chaufita,	0.478
… la teletón es una estafa o entiendo chaparrón	chapa	chaparrón	0.565
	chapar	chaparrón	0.738
… CONERAS los alaracos de mi barrio 🧐 BUENO ESO ENTENDÍ XD 1ra vez ENTENDÍ bien hablar los chilenos. SALUDOS	habla	hablar	0.734
	conera	CONERAS	0.877
Melanie Arce Rodriguez mira amor que capo! 🤣	capo	capo!	0.858
O chino pagales con un turrón y a la firme que feo mostró …. 🙌	la firme	firme	0.790
	chino	chino	1.000
	firme	firme	1.000
	turrón	turrón	1.000



Figure 2. N-Gram of the slang in the dataset.

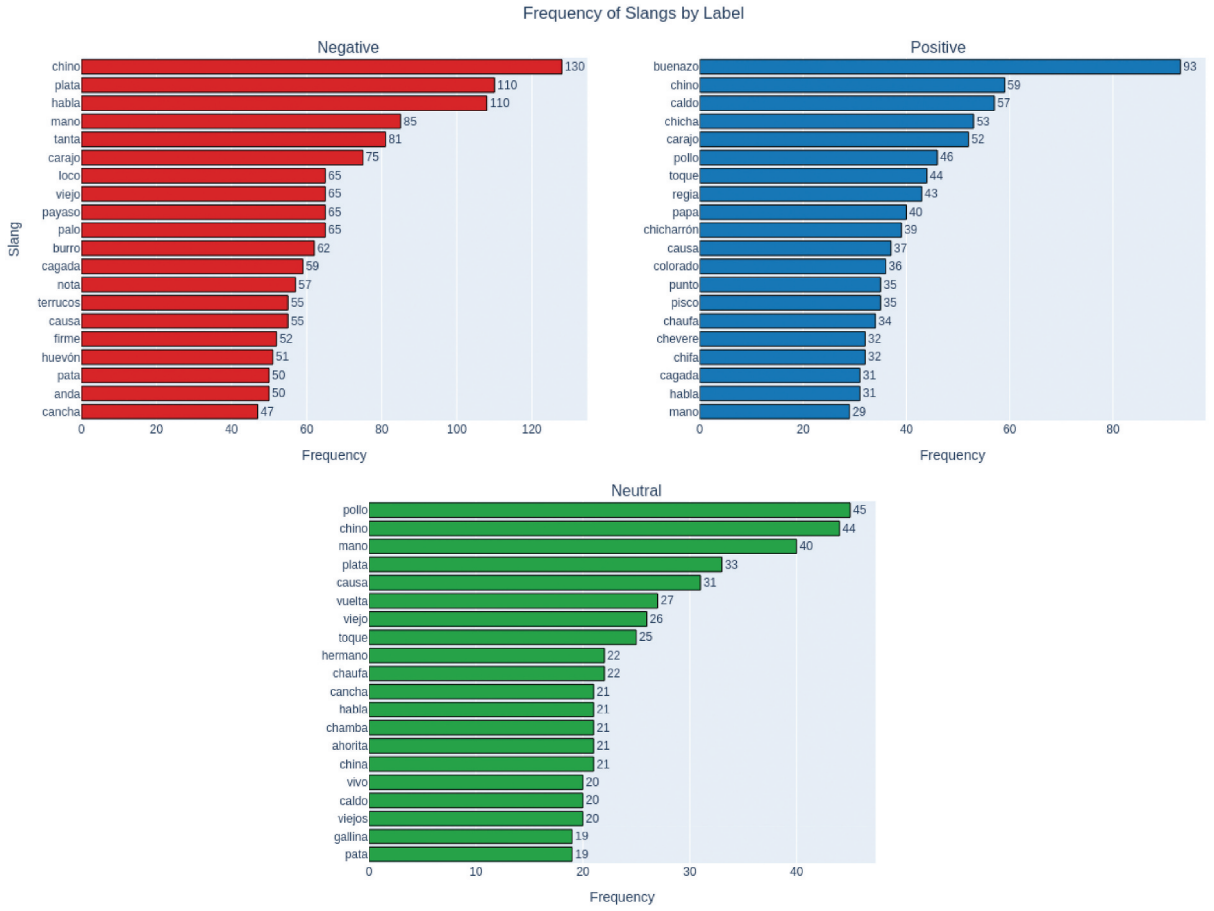


Figure 3. Frequency of slang in each positive, negative, and neutral class.

- Positive: Comments classified as positive expressed favorable opinions, satisfaction, or praise toward a person, event, or idea. In the context of Peruvian slang, colloquial terms that commonly denote appreciation or admiration, such as "capo," "buenazo," and "trome," were included. These terms are typically used to highlight positive qualities in a person, as in "Buena Carlitos, eres un capo" (meaning someone is very talented or outstanding). Additionally, comments that, even in a humorous or joking tone, did not contain insults or ill intent (banter) were also labeled positive.
- Negative: Negative comments reflected disapproval, criticism, or dissatisfaction. In many cases, insults characteristic of urban culture, mockery, or high use of sarcasm were identified. Terms like "palta" (referring to an embarrassing or problematic situation) were commonly labeled as negative, as comments that expressed dissatisfaction or rejection, either explicitly or through irony. These comments could include direct language or imply discontent through specific cultural connotations.

Table 3. Example of labeled comments from the SMPS dataset.

ID	Comments	Slang	Polarity
2037	Buena carlitos! Eres un capo🤣 🤣.	capo	Positive
7041	Que éxito!unos tromes todos !no hay como nuestro chifa! Es delicioso👌👌👌👌👌👌.	trome	Positive
3810	Mira Camilo Ahi tienes que empujarte tu cevillano.	cevillano	Neutral
19964	Con solo ver este video de comida ya me dio la bajada.	bajada	Neutral
18615	Jajajajajajaja 🙂 ahora ya quiere pasar piola para que no la sigan atacando 😂.	piola	Negative
244536	😤😠causa! Tú con todas las publicaciones q realizas y más palta los gobiernos que hemos elegido hasta antes de Vizcarra🥑🥑🥑🥑🥑🤦🏾♂️🤦🏾♂️.	palta, causa	Negative

Table 4. Polarity distribution in the labeled dataset.

Polarities	Number of comments	%
Positive	4,736	42%
Negative	3,280	29%
Neutral	3,260	29%
Total	**11,276**	**100%**

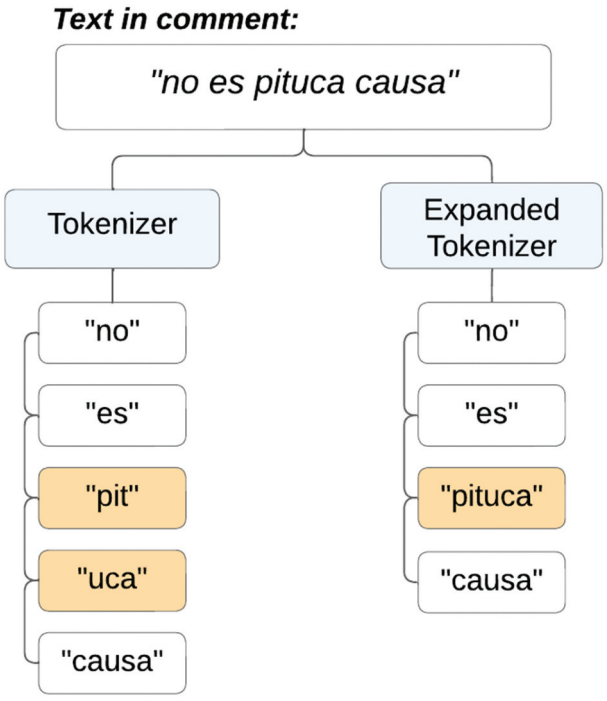


Figure 4. Word tokenization.

- Neutral: Comments were labeled neutral when they did not show a clear positive or negative opinion. Usually, these comments consisted of descriptions or information without significant emotional charge. Regarding slang, neutral terms were those that did not convey a specific emotion or were used descriptively, such as in "Su chapa es el gordo" (his nickname is "el gordo") or "Con solo ver este video de comida ya me dio la bajada" (just watching this food video made me hungry), where the words "gordo" and "bajada" did not contain judgment or offensive intent. Some slang, like "pulpín" (young person), was particularly difficult to classify due to its ambiguity, as its interpretation depends on the context. For example, in "Ese pulpín, ¿a dónde se fue?" (Where did that young person go?), "pulpín" can be neutral if no judgment is expressed about the person's age.

There was a need to manage samples that contained only emojis rigorously, comments that were too short or long and lacked meaning, spam or advertising messages, intentionally distorted abbreviations, or user mentions without relevant content. In these cases, annotators were instructed to evaluate the overall intent of the comment and, if deemed meaningless or irrelevant, classify it as "invalid." These invalid comments were removed from the SMPS dataset used for training the models, thus ensuring the quality and consistency of the corpus.

The most frequent slang in the analysis is "chino," a colloquial term commonly used to refer to a friend or acquaintance or as a nickname. The frequency of the most common slang terms across the positive, negative, and neutral classes is shown in Figure 3. In some cases, it can also refer to facial features associated with Asian ancestry. This term appears in all three polarity categories, indicating its semantic ambiguity and ability to acquire different interpretations depending on the context. Similarly, the word "carajo" is predominantly used in the positive and negative categories, and its connotation varies according to usage. In certain contexts, "carajo" can express frustration or anger, as in "¡No puede ser que hayan perdido

el partido otra vez, carajo!" (I cannot believe they lost the game again, damn it!). In other cases, it can express enthusiasm or affection, as in "¡Viva el Perú, carajo!" (Long live Peru, damn it!). On the other hand, the word "buenazo" is the most frequent in the positive category, reflecting its use in situations of approval or praise.

Tokenization process

The word tokenization technique was implemented to break down the texts into minimal units or individual words. During this process, it was observed that the predefined tokenizers of the language models did not adequately recognize the Peruvian slang present in the dataset. A clear example of this deficiency can be seen in the phrase "no es pituca causa," where the models' tokenizer did not index the word "pituca" or other Peruvian slang.

To overcome these limitations, the models' base vocabulary, which initially consisted of 30,000 tokens, was manually expanded. This process involved incorporating new tokens extracted from the Peruvian slang dictionary, resulting in an expanded vocabulary of 31,886 tokens. This expansion allowed for a more accurate breakdown of texts that included slang or new words, significantly improving the data preprocessing quality before the training phase. Figure 4 illustrates this improvement.

Model training

To adapt the pre-trained Spanish models RoBERTuito, BETO, BERTuit, and RoBERTa-BNE to our specific text classification task, we performed parameter tuning of the models. This process included setting a learning rate of (5e-5), following the guidelines of (Devlin et al. 2019), which suggest a low rate for gradual and effective changes in the model's pre-trained weights, a train batch size of 32, and an evaluation batch size of 16. The tuning process was carried out over a total of 5 epochs, which was considered sufficient to allow the models to capture the specific task's particularities without losing the richness of the linguistic representations acquired during the pre-training phase. This approach also helps mitigate the risk of overfitting. A warm-up of 10% of the training steps was implemented, allowing for a smooth start to the optimization process.

Each model's embeddings were resized to 31,886 according to the specific vocabulary of its tokenizer, ensuring an accurate representation of Peruvian slang. For the data split, 80% of the total set was used for training, and the remaining 20% was allocated for testing and validation.

The models were trained using a computer equipped with an NVIDIA RTX 3060 card with 12 GB of RAM and an Nvidia T4 GPU instance in a Google Colab environment. The training time per model ranged from 15 to 20 min, demonstrating that fine-tuning pre-trained monolingual models focused on specific dialects and slang can be done efficiently without requiring excessively high computational resources.

The architecture of the RoBERTuito, BETO, BERTuit, and RoBERTa-BNE models belongs to the Transformer family of models, initially introduced by (Vaswani et al. 2017). This architecture allows the models to process extensive text sequences. Thanks to its self-attention and multi-headed attention mechanisms, the models can capture complex relationships within the text. Specifically, the RoBERTuito model, based on the RoBERTa (Liu et al. 2019) and BERTweet (Nguyen, Vu, and Tuan 2020) architectures, has been trained with 12 self-attention layers, each with 12 attention heads and a hidden size of 768 dimensions. For RoBERTuito, we used the *deacc* version, where the model was trained in lowercase and without accents.

Evaluation and results

For the evaluation of the sentiment analysis classification models, standard metrics such as precision, recall, accuracy, F1 Score, and Macro F1 Score were applied (Murphy 2012). These metrics provide a comprehensive view of the models' effectiveness in correctly classifying the analyzed comments. Accuracy indicates the proportion of overall correct predictions, meaning the total number of cases the model correctly classifies, considering both positives and negatives. On the other hand, precision measures how accurate the positive predictions are, indicating the proportion of cases correctly identified as positive

compared to the total number of predictions made. Meanwhile, recall evaluates the model's ability to identify all positive cases, representing the proportion of positive cases that were correctly predicted in relation to the total number of positive cases. The F1 Score provides a balanced measure that considers both precision and recall, calculated as the harmonic mean of both metrics (Hidayat et al. 2021).

$$Accuracy = \frac{tp + tn}{tp + tn + fp + fn} \tag{3}$$

$$Precision = \frac{tp}{tp + fp} \tag{4}$$

$$Recall = \frac{tp}{tp + fn} \tag{5}$$

$$F1Score = 2 \cdot \frac{precision \times recall}{precision + recall} \tag{6}$$

$$MacroF1 = \frac{1}{N}\sum_{i=1}^{N} F1_i \tag{7}$$

Where *tp* represents the number of true positives, meaning cases where the model has correctly identified a comment as positive; *tn* denotes the number of true negatives, or negative comments correctly classified; *fp* corresponds to false positives, where a negative comment is incorrectly identified as positive; and *fn* reflects false negatives, situations in which positive comments are wrongly classified as negative. The Macro F1 Score, in turn, calculates the arithmetic mean of the F1 Scores obtained for each of the model's classes, providing a balanced overview that is not influenced by class imbalance. In Table 5, we can observe that RoBERTuito shows the most outstanding performance regarding precision and recall for both positive and negative comments. Specifically, RoBERTuito achieves a precision of 0.858 for positive comments and 0.828 for negative ones, with a recall of 0.870 and 0.798, respectively. These results indicate that RoBERTuito is highly effective in correctly identifying positive and negative comments, with an F1 Score of 0.864 for positive and 0.813 for negative comments.

BETO and BERTuit also show promising results. BETO achieves an F1 Score of 0.758 for positive comments and 0.685 for negative ones, with a precision of 0.794 for positives and 0.701 for negatives. BERTuit obtains an F1 Score of 0.806 for positives and 0.741 for negatives, with a precision of 0.751 and 0.746, respectively. These models demonstrate their ability to correctly classify comments, though with slightly lower performance than RoBERTuito.

The RoBERTa-BNE model presents balanced performance, with a precision of 0.783 for positive comments, 0.746 for negative ones, and a recall of 0.759 and 0.740, respectively. Its F1 Score of 0.771 for positives and 0.743 for negatives indicates a good balance between precision and recall in these categories.

It has been noted that all models experience difficulties in classifying neutral comments, with significantly lower precision and recall scores in this category. RoBERTa-BNE performs best in neutral comments among the evaluated models, with an F1 Score of 0.575. RoBERTuito and BERTuit follow with F1 Scores of 0.572 and 0.552, respectively. BETO also performs moderately in this category, with an F1 Score of 0.542. These challenges suggest further refinement of the models to improve the classification of neutral comments.

Table 5. Performance evaluation results for the fine-tuned models.

	RoBERTuito			BETO			BERTuit			RoBERTa-BNE		
Metrics	P	Neu	N	P	Neu	N	P	Neu	N	P	Neu	N
Precision	0.858	0.565	0.828	0.794	0.502	0.701	0.751	0.629	0.746	0.783	0.561	0.746
Recall	0.870	0.578	0.798	0.725	0.588	0.669	0.869	0.492	0.736	0.759	0.590	0.740
F1-Score	0.864	0.572	0.813	0.758	0.542	0.685	0.806	0.552	0.741	0.771	0.575	0.743
Accuracy	0.789			0.669			0.722			0.750		
Macro F1	0.750			0.661			0.700			0.696		

The confusion matrices presented in Figure 5 show the performance of the RoBERTuito, BERTuit, RoBERTa-BNE, and BETO models in classifying comments into three categories: negative, neutral, and positive. The RoBERTuito model demonstrates solid ability in classifying negative comments (31.10%) and positive ones (21.17%), although, like the other models, it faces challenges in identifying neutral comments, with a 13.31% success rate and a tendency to confuse negative comments with neutral ones (14.04%).

On the other hand, BERTuit performs best in classifying negative comments, with 39.28% accuracy, but also higher confusion when classifying neutral comments as negative (7.07%) and positive (22.02%). This indicates that BERTuit tends to classify more extremely but maintains good accuracy at the extremes of polarity.

RoBERTa-BNE, though similar in performance to BERTuit in classifying positive comments (22.97%), shows a lower ability to identify neutral comments, with a 12.04% success rate and significant confusion in classifying negative comments as neutral (9.61%). This model strikes a balance in classifying negative and positive comments but struggles with accurately identifying neutrality.

Finally, BETO, with a notable performance in classifying negative comments (42.82%), shows the highest confusion in the neutral category, with an 8.76% success rate and a considerable number of false positives

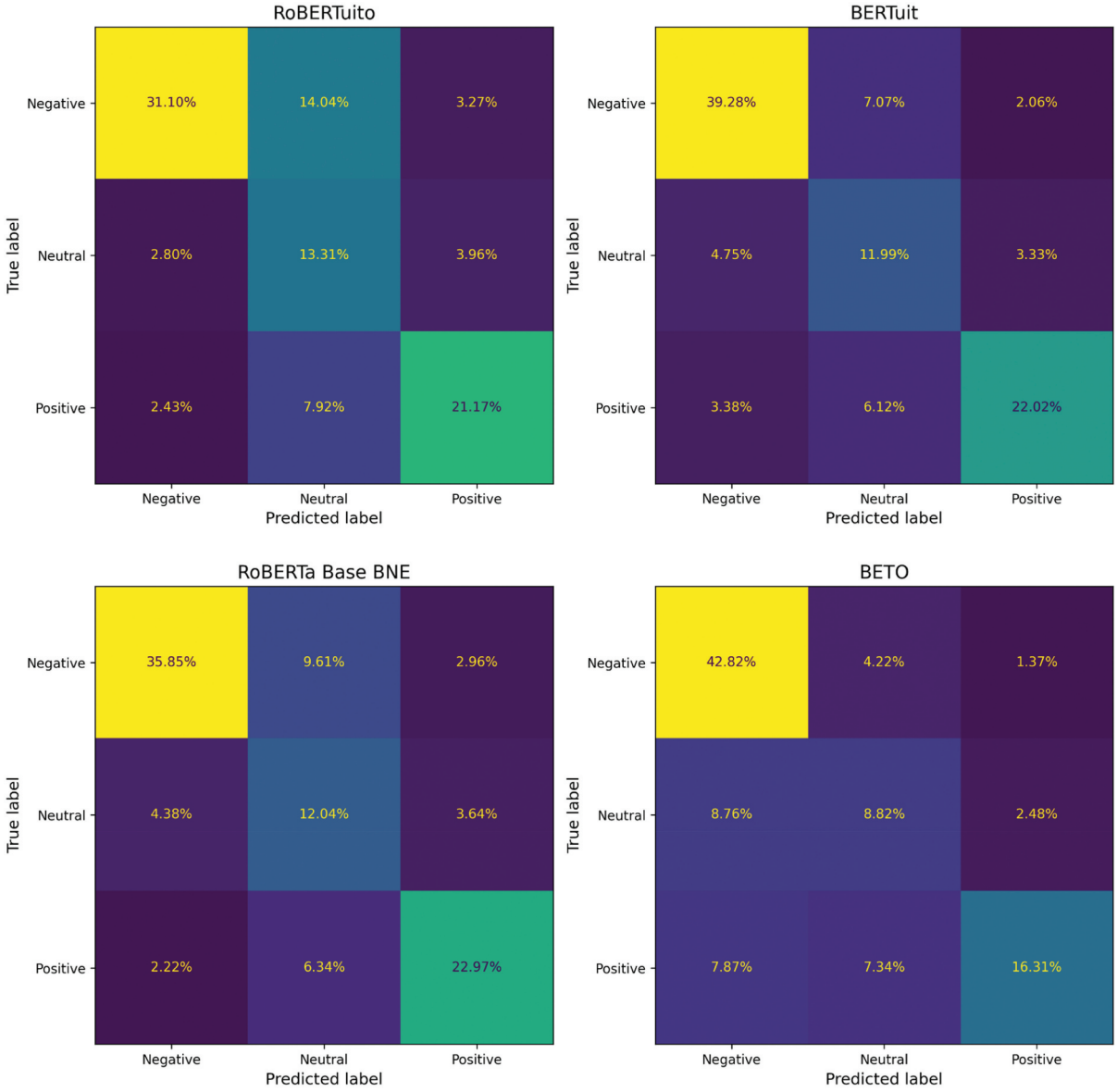


Figure 5. Confusion matrices of the fine-tuned models.

Table 6. Evaluating fine-tuned BERT models in text classification, bold numbers indicate high scores.

Models	Macro F1	Macro F1 (Fine-Tuned)
RoBERTuito	0.707	**0.750**
BETO	0.665	0.661
BERTuit	0.632	**0.700**
RoBERTa-BNE	0.669	0.696

and negatives. BETO also has the lowest accuracy in identifying positive comments (16.31%), indicating that this model struggles to capture the positive context correctly.

The overall results shown in Table 6, compared to the results reported in the text classification tasks of each respective model (Pérez et al., 2022) and (Huertas-Tato, Martin, and Camacho 2023), indicate that the model with the best performance in polarity classification is RoBERTuito, achieving a Macro F1 of 0.750, followed by BERTuit, which shows a score of 0.700. These two models demonstrate great capability in text classification within the dataset. On the other hand, BETO showed a respectable performance with a score of 0.661. The RoBERTa-BNE model also presents solid performance with a Macro F1 of 0.696, indicating that, while slightly lower than BERTuit and RoBERTuito, it remains effective in polarity classification.

Discussion

This research highlights the challenging nature of sentiment classification in colloquial texts and the importance of considering linguistic and cultural nuances in NLP models. Despite the advancements achieved by models like RoBERTuito, BETO, and BERTuit in processing standard Spanish texts, challenges remain when adapting them to dialectal variations, as evidenced by our analysis of Peruvian slang and its results. In our study, we employed Transformer-based models and obtained promising results. Consistent with our observations, the study by (Merayo et al. 2024) demonstrated significant improvements when applying a Transformer-based model compared to conventional methods, specifically RoBERTuito, in an informal language context. Therefore, a more detailed and contextualized approach is suggested for future research.

In contrast to previous studies, our research emphasizes the importance of a well-defined, specific dataset, which does not need to be extensive, to improve sentiment classification. Unlike general approaches that may fail to capture the idiosyncrasies of colloquial language and its context, our dialect-focused approach shows that polarity classification accuracy can improve through fine-tuning pre-trained monolingual models. A clear example is the study by (Kannan and Kothamasu 2022), where they achieved an average of up to 25% more performance in their fine-tuned model compared to a base BERT model. Similarly, Fernandez et al. (2022) achieved an accuracy of 60.35% in a similar colloquial language context by fine-tuning IndoBERT, a version of BERT for the Indonesian language, with a relatively small corpus. In contrast, we achieved an accuracy of 78.9%. While it is true that we are not comparing the same language, both studies address an informal language context and yield promising results. This reinforces the idea that fine-tuning monolingual models does not necessarily require large volumes of text to achieve outstanding results. It is essential to consider the complexity of slang and irony because these elements rely heavily on context rather than the literal meaning of words, as noted by (González, Hurtado, and Pla 2020). This opens new possibilities to explore how advanced linguistic models, combined with more precise annotation techniques, can better address the challenges posed by ambiguous or emotionally confusing comments.

Conclusions and future work

This research addresses the challenge of classifying texts that contain slang specific to Peruvian culture, using a manually labeled dataset with polarity annotations and trained with monolingual models for the Spanish language. The model with the best performance was RoBERTuito, achieving a Macro F1 Score of 0.750 after fine-tuning. This fine-tuned model is referred to as RoBERTuito.pe.[7] This finding demonstrates that fine-tuning pre-trained models in Spanish with lexicons specific to Peruvian Spanish can significantly improve text classification. However, areas for improvement were identified, particularly in the

classification of neutral comments. This indicates the need for further model refinement by exploring other advanced NLP techniques to better handle ambiguities and context in this data type.

Throughout this study, it has been confirmed that the variety and integrity of the dataset influence the results. Consequently, we plan to expand data collection to include more current colloquial expressions and conduct additional experiments to refine the findings. The developed dataset[8] is available for use and free download from the Hugging Face platform, and we have made both the text corpus[9] and the scripts used throughout the process are available in a GitHub repository.[10] This facilitates its use in future research for other tasks addressing Peruvian Spanish, such as emotion classification or challenges that persist in NLP, particularly in specific dialects like irony, sarcasm, or hate speech.

Moreover, our study's extensive variety of comments presented challenges in establishing annotator agreement. The agreement index was not directly calculated since the annotators worked randomly on over 11,000 comments. Therefore, we propose conducting a detailed analysis of the agreement among three or more annotators. This effort will not only validate the consistency of the assigned labels but also refine the accuracy of our annotated dataset. Finally, we plan to compare the performance of monolingual models against the latest LLMs in the task of text classification using the proposed dataset.

Notes

1. https://github.com/strohne/Facepager.
2. https://www.asale.org/damer/.
3. https://huggingface.co/datasets/pyupeu/peruvian-slangs-dictionary.
4. https://github.com/carpedm20/emoji/.
5. https://github.com/doccano/doccano.
6. https://huggingface.co/datasets/pyupeu/social-media-peruvian-sentiment.
7. https://huggingface.co/pyupeu/robertuito-peruvian-sentiment.
8. https://huggingface.co/datasets/pyupeu/social-media-peruvian-sentiment.
9. https://huggingface.co/datasets/pyupeu/social-media-peruvian-corpus.
10. https://github.com/jairleo95/peruvian-slang-sentiment.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data supporting the results of this study are publicly available on Figshare at http://doi.org/10.6084/m9.figshare.26198315., re ference number 26,198,315.

Ethics statement

This study involved the analysis of text comments collected from content that Meta (Facebook) describes as public data on its platform and in its developer documentation. The study received retrospective confirmation from the university's ethics committee (Approval No. 2025-CEUPeU-027; 16 June 2025) that the data collected conforms to Meta's definition of public data and applicable platform terms. Informed consent was exempted because the dataset does not contain information that identifies users, in accordance with Meta's guidelines and terms. All data were processed and stored in de-identified form and used exclusively for research purposes.

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